

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

- **Question:** Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Solution:- `CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest AS
BEGIN UPDATE Accounts
SET Balance = Balance + (Balance * 0.01) WHERE AccountType = 'Savings';`

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

- **Question:** Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Solution:`CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(p_DepartmentID IN
NUMBER, p_BonusPercent IN NUMBER) AS
BEGIN UPDATE Employees
SET Salary = Salary + (Salary * p_BonusPercent / 100) WHERE DepartmentID =
p_DepartmentID;`

Scenario 3: Customers should be able to transfer funds between their accounts.

- **Question:** Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Solution:-`CREATE OR REPLACE PROCEDURE TransferFunds(p_FromAccount IN NUMBER,
p_ToAccount IN NUMBER, p_Amount IN NUMBER) AS v_Balance NUMBER;
BEGIN SELECT Balance INTO v_Balance FROM Accounts WHERE
AccountID = p_FromAccount;
IF v_Balance >= p_Amount THEN UPDATE Accounts
SET Balance = Balance - p_Amount WHERE AccountID = p_FromAccount;
UPDATE Accounts SET Balance = Balance + p_Amount
WHERE AccountID = p_ToAccount;`